

e-ink electronic paper works by using electrodes to arrange thousands of tiny black and white charged capsules into shapes to form characters. Changing the charge changes the characters the electronic paper displays. The intelligent ink uses power only when the price is changed, so the battery lasts for ages. This technology can free up staff time considerably.

Solar gadgets, from the serious to the silly

Save energy with the Solar Ski jacket by Zegna Sport. This jacket has solar panels on the removable collar that convert sunlight into energy to directly charge up gadgets such as iPods and cell phones or a rechargeable battery. An hour of direct sunlight will charge up a phone for about 35 minutes of talk time. The battery requires four to five hours for a full charge. The collar and battery are detachable for use even when you're inside by the fire. The jacket is priced at \$1350.

The Solio Classic portable charger also comes in handy for gadget geeks. When fanned open, it soaks up energy from the sun. Adaptor tips send the



Solar Ski jacket by Zegna Sport

charge to everything from iPods to digital cameras and GPS devices. Eight to 10 hours of direct sunlight charge the gizmo's internal battery. The charger is priced at \$99.95.

Light to power nanoscale machines?

You can't feel it, but light can exert forces. And it has now been used to drive a tiny mechanical resonator, opening the door to a new way of powering nanoscale machinery.

The device was developed by Hong Tang's team at Yale University. It contains a waveguide—a kind of constricting channel for light—which sends light through a bottleneck in a silicon photonic circuit. That 10µm long, 110nm wide channel constrains the light, causing the material to resonate at right angles to the beam. This vibration is the result of the so-called optical gradient force, which kicks out sideways from the light beam. Because light can be easily beamed onto large areas, many such photonic circuits could be driven at once.

The Yale team's work is the first to demonstrate that those forces can be usefully collected in even smaller, nanoscale circuits. They say their photonic circuits could be used to control other small-scale devices, such as electronics or light-based computer chips.

See-through touch-screen

It is often assumed that touch-screens make handheld devices smaller, as these do away with the need for buttons. But paradoxically, they are often larger than the pushbutton gadgets they replace, according to Patrick Baudisch at the Hasso Plattner Institute in Potsdam, Germany, and Microsoft Research in Redmond, Washington. It is hard to hit small touch targets on a screen because your finger hides what you are aiming for.

Baudisch and his student Gerry Chu at the University of Toronto



NanoTouch screen with a touchpad of the same size on the back

have developed a fix to that. Called NanoTouch, it has a 6cm screen and a touch pad of the same size on the back. It can detect the touch or press of a finger, allowing you to move a tiny cursor around and click and drag with it.

When you touch the interface on the back, an image of a finger appears behind the icons on the screen and moves around in sync with your finger, almost as if the device were transparent. A small active spot marked on the finger's end is used to interact with buttons onscreen. User tests showed that targets just 1.8 mm across were easy to hit using NanoTouch. Targets on conventional touch-screen devices such as the iPhone are at least twice that size.

Smallest humanoid robot

The world's smallest humanoid robot, i-Sobot, has a big personality all of its own. It measures only 16 cm, i-Sobot robot weighs an amazing 300 grams and comes ready to play straight from the box. You can choose from four modes where i-Sobot can perform fun and quirky scenes from Western battles to voice command mode that includes ten words of phrases to say to i-Sobot where he will respond.

Thanks to 17 actuators, a gyroscope and a whole bunch of geeky

